THE MASTER QUADRATIC P_{G^*} AS A MATH–PHYSICS BRIDGE: χ_{-4}, THE TERNARY SUBSTRATE, AND THE FTD IDENTIFICATION

ABSTRACT. The companion paper [1] establishes that the Kronecker character χ_{-4} of $K = \mathbb{Q}(i)$ generates the entire algebraic–analytic identity ring of the lemniscatic constant $G^* = \Gamma(1/4)/\Gamma(3/4)$ through four functorial projections (lattice, Chowla–Selberg, Hecke, Dirichlet), and that $\chi_{-4}(n) = \mathrm{Im}(i^n)$ with value set $\{-1, 0, +1\}$.

This companion note draws the consequences of those results for the math–physics bridge of the Foundational Ternary Dynamics (FTD) framework [2], where the same set $\{-1, 0, +1\}$ appears as the ternary voxel alphabet of a discrete physical substrate. Three structural observations are recorded, all explicitly tagged as synthesis rather than theorem:

(i) the FTD voxel alphabet coincides as a set with the value set of χ_{-4}, sourcing a candidate *Gaussian-integer-module ontology*;

(ii) the Born rule $\psi \mapsto |\psi|^2$ and the character $\chi_{-4} : i^n \mapsto \mathrm{Im}(i^n)$ are two instances of a general "complex object $\to$ single real coordinate" schema, sharing ternary or near-ternary output but differing in domain, codomain, and coordinate read;

(iii) the FTD identification $(x_+, x_-) \approx (\alpha^{-1}, N_c)$ of the roots of the master quadratic $P_{G^*}(x) = x^2 - 16G^{*2}x + 16G^{*3}$ with the fine-structure constant and the number of QCD colours requires a single discrete integer-pair selector — the exponent pair $(2, 3)$ — which is over-determined by joint root-matching.

All mathematical content is taken from [1]; this companion contributes only the bridge interpretation, presented under the project's `[SYNTHESIS]` tag.

1. SETTING AND SCOPE

We adopt the entire mathematical content of [1] without re-derivation. The constants in use are:

- $G^* = \Gamma(1/4)/\Gamma(3/4) \approx 2.9587$;
- $G_{\mathrm{G}} = 1/\mathrm{AGM}(1, \sqrt{2}) \approx 0.8346$, with the bridge identity $G^* = 2\sqrt{\pi}\, G_{\mathrm{G}}$;
- $\chi_{-4} : (\mathbb{Z}/4\mathbb{Z})^\times \to \{\pm 1\}$, the unique non-trivial Dirichlet character mod 4;
- the master quadratic $P_{G^*}(x) = x^2 - 16\, G^{*2}\, x + 16\, G^{*3}$, with integer coefficient $16 = |\mathrm{Aut}_{\mathbb{C}}(E_{\mathrm{lemn}})|^2$ and roots $x_\pm$ satisfying Vieta's identities $x_+ + x_- = 16G^{*2}$, $x_+ x_- = 16G^{*3}$, $1/x_+ + 1/x_- = 1/G^*$.

The mathematical paper [1] proves that χ_{-4} admits the explicit form $\chi_{-4}(n) = \mathrm{Im}(i^n) = \sin(\pi n/2)$ and that its value set is

$$\{\chi_{-4}(n) : n \in \mathbb{Z}\} \;=\; \{i^2,\, 0,\, |i^2|\} \;=\; \{-1, 0, +1\}.$$

Numerical observation: the roots of P_{G^*} are $x_+ \approx 137.036\ldots$ (matching the inverse fine-structure constant α^{-1} to 1.26 ppm) and $x_- \approx 3.024\ldots$ (matching the number of QCD colours $N_c = 3$ to 0.8%). The Foundational Ternary Dynamics framework [2] posits the identification $(x_+, x_-) = (\alpha^{-1}, N_c)$ as a strongly motivated conjecture.

This note is the math–physics bridge: it draws structural consequences of the χ_{-4}-unification of [1] for the FTD identification, without proving the identification itself. All load-bearing claims are tagged `[SYNTHESIS]` or `[STRONGLY MOTIVATED CONJECTURE]` as appropriate; no new mathematical theorem is claimed in this note.

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2. The value-set coincidence and the Gaussian-integer-module ontology

Observation (Value-set coincidence). **[Synthesis.]** Three independent starting points terminate at the same three-element set:

(1) the FTD physical primitive — each lattice site v takes a state $s_v \in \{-1, 0, +1\}$ encoding *negative state*, *void*, *positive state* [2];
(2) the value set of the unique non-trivial real Dirichlet character of conductor 4 [1, §16];
(3) the closure of the imaginary-unit orbit $\{i^n : n \in \mathbb{Z}\}$ under the imaginary-part map: $\{\mathrm{Im}(i^0), \mathrm{Im}(i^1), \mathrm{Im}(i^2), \mathrm{Im}(i^{3}$ $\{0, 1, 0, -1\}$.

That these three sets coincide is a set-level identity, not yet a *functorial* identification. Whether the FTD voxel alphabet *arises from* the character χ_{-4}, or only happens to share its value set, is a physical-modelling question that lies outside the scope of the mathematical paper.

Remark (Candidate ontology: Z[i]. -substrate realism] **[Synthesis.]** The value-set coincidence of Observation 2 suggests a working ontology in which the empirical universe is the real-coordinate readout of a $\mathbb{Z}[i]$-module structure. Layering the established FTD framework against this ontology:

- voxels are positions in a $\mathbb{Z}[i]^d$-style lattice ($d = 3$ in current FTD);
- voxel states $s_v \in \{-1, 0, +1\}$ are the values $\mathrm{Im}(i^\bullet)$ readouts of the unit group $\mathbb{Z}[i]^\times$;
- the flux field $J \in \mathbb{R}^3$ is the continuous real part of an underlying complex object;
- time is the order parameter of the cyclic action $\langle i \rangle$ at long wavelength;
- the mass–length calibration $a_{\text{phys}} \equiv \ell_P$ is the choice of unit on the $\mathbb{Z}[i]$-module;
- dimensionless physical constants (α, N_c, mass ratios) are intrinsic invariants of the $\mathbb{Z}[i]$-module structure.

The ontological position is most accurately described as *Pythagorean structural realism, specialised to the Gaussian-integer module*: Pythagorean because number is taken as the substance from which the physical emerges; structural realist in attitude (entities are nodes in the structural network rather than independent objects — compare [4], though our specialisation to a single algebraic structure goes beyond their naturalistic stance); specialised because the particular structure is the maximal order of $\mathbb{Q}(i)$ rather than an arbitrary mathematical structure. This is a more specific, and therefore more falsifiable, restriction of an MUH-style position [3]: not "all mathematical structures are physical" but "the physical is the $\mathbb{Z}[i]$-module slice of the mathematical." Under this specialisation Wigner's "unreasonable effectiveness" [5] becomes a definitional consequence rather than a mystery.

Remark (Falsifiability). **[Synthesis.]** The $\mathbb{Z}[i]$-substrate ontology is empirically refutable: if experiment ever produced an observable that requires a $\mathbb{Z}[\rho]$-module structure (Eisenstein, sixfold) that cannot be embedded in a $\mathbb{Z}[i]$ ambient, or that requires $\mathbb{Z}[\sqrt{-2}]$ or $\mathbb{Z}[\sqrt{-7}]$ (other class-number-1 quadratic orders), the framework breaks. The position has therefore the falsifiability profile of a substantive empirical hypothesis, not a metaphysical claim.

3. The Born rule and χ_{-4}: a schema and a conjecture

The χ_{-4}-unification of [1] makes available a structural schema that connects two otherwise unrelated operations.

Observation (Born rule and χ_{-4} as two instances of a real-coordinate schema). **[Synthesis.]** The Born rule of quantum mechanics and the character χ_{-4} are two instances of a general schema "complex object $\to$ single real coordinate $\to \mathbb{R}$," chosen independently in QM and in number theory for unrelated structural reasons:

- **Born rule**: $\psi \in \mathbb{C} \ \mapsto \ |\psi|^2 \in \mathbb{R}_{\geq 0}$, the squared modulus, a continuous map whose load-bearing content is the Gleason-theorem link between an inner-product-space amplitude and a probability measure.
- **Character** χ_{-4}: $i^n \in \langle i \rangle \subset \mathbb{C} \ \mapsto \ \mathrm{Im}(i^n) \in \{-1, 0, +1\}$, the imaginary-part trace on a finite cyclic group, a discrete map whose content is the detection of quadratic residues mod 4.

The two operations differ in their domain (continuous Hilbert space vs. finite cyclic group), their codomain (positive half-line vs. signed three-point set), the coordinate they read (modulus vs. imaginary part), and their non-linearity (squared modulus vs. linear imaginary part). The observation worth recording is therefore not that they are "the same arrow," but that both produce ternary or near-ternary outcomes when restricted to relevant subdomains, by virtue of the imaginary-unit orbit $\langle i \rangle$ having exactly three distinct imaginary projections.

Remark (Reflexive projection at two identical scopes and one analogous scope). **[Synthesis.]** Under the FTD framework's identification of the voxel state field with the substrate's reflexive projection [2], two scopes are *literally the same operation* and a third is *conjecturally analogous*:

- (Identical) The FTD voxel state field as the imaginary-part readout of $\mathbb{Z}[i]^\times \cup \{0\}$, and the character χ_{-4} as $\mathrm{Im} \circ (i^\bullet)$ on the angular index, are literally the same map; their coincidence is a definitional consequence of the $\mathbb{Z}[i]$-substrate position rather than an independent identification.
- (Analogous) The Born rule of quantum mechanics is conjecturally related to this same map at a derived scope (amplitude measurement of $\mathbb{Z}[i]$-structured quantum states). This is the load-bearing physical conjecture, tagged `[STRONGLY MOTIVATED CONJECTURE]`; it is not established by any mathematical theorem and the differences enumerated in Observation 3 mean it should not be conflated with the two identical scopes.

4. The master quadratic as the bridge

The FTD bridge from χ_{-4}-derived mathematics to physical observables runs through the master quadratic P_{G^*}. We recall (from [1, Theorem 16.2]) that χ_{-4} generates the integer coefficient $16 = |\mathbb{Z}[i]^\times|^2$ (lattice projection) and the constant G^* (Chowla–Selberg projection), but does *not* generate the specific polynomial form $x^2 - 16G^{*2}x + 16G^{*3}$ from classical CM/Weber/Hecke machinery (the three negative tests of [1, §16.5]).

The exponent pair $(2, 3)$ on G^* in the polynomial coefficients is nevertheless *uniquely picked out* by joint root-matching:

Observation (Joint-matching uniqueness, restated from [1, §16.6]). **[Synthesis.]** Among polynomial forms $y^2 - 16R^p y + 16R^q$ at $R = G^*$ with $(p, q) \in \mathbb{Z}^2$, $|p|, |q| \leq 5$, only the pair $(p, q) = (2, 3)$ produces an ordered root pair (y_+, y_-) that simultaneously matches the FTD pair (α^{-1}, N_c) to sub-percent precision. The match precision at $(2, 3)$ is 1.26 ppm for $y_+ = 137.036$ and 0.8% for $y_- = 3.024$. Other exponent pairs match one root but not both, or neither.

Remark (Information-theoretic minimality of the FTD bridge). **[Synthesis.]** The FTD math–physics bridge requires exactly *a single discrete integer-pair selector* from the physical side — the exponent-pair selector $(p, q) \in \{(2, 3), (3, 4), (4, 5), \ldots, (1, 2), (2, 4), \ldots\}$ — and that bit is over-determined by joint matching to two distinct physical constants (α^{-1}, N_c). Equivalently: the FTD master quadratic is the unique polynomial of the form $y^2 - 16R^p y + 16R^q$ at $R = G^*$ whose ordered root pair (y_+, y_-) jointly matches (α^{-1}, N_c) to sub-percent precision.

This is not a derivation of the bridge — the joint match itself is the input that selects (p, q) — but it is the cleanest available *minimality statement*: the math–physics bridge in the FTD framework requires no parameters beyond the choice of one exponent pair, and the choice is over-determined (hence robust) against the joint match to two distinct physical constants.

Observation (The locus of the math–physics gap). Combining [1, §16.5] (no CM-internal source for $(2,3)$) with Observation 4 (joint match uniquely fixes $(2,3)$): the gap between χ_{-4}-derived mathematical structure and the FTD physical identification is localised precisely at the polynomial *assembly* step. The integer coefficient 16 and the constant G^* are both forced by χ_{-4}. The exponent pair $(2,3)$ is forced only by the joint physical-matching requirement. Closing this last gap — if a clean derivation of $(2,3)$ from χ_{-4}-internal data exists — would convert the FTD bridge from a two-arrow conjecture into a single-arrow theorem.

5. The full descent chain

The cumulative content of [1] and this companion can be displayed as a single descent chain from physics to ontological bedrock:

$$\underbrace{(\alpha^{-1},\, N_c)}_{\text{measured physical constants}}$$

$$\Big\uparrow \quad \texttt{[STRONGLY MOTIVATED CONJECTURE]} \quad \text{(FTD bridge)}$$

$$\underbrace{P_{G^*}(x) = x^2 - 16{G^*}^2 x + 16{G^*}^3}_{\text{master quadratic}}$$

$$\Big\uparrow \quad \text{(2, 3) exponents: uniquely forced by joint root match [1, §16.6]}$$

$$\underbrace{\{16,\ G^*\}}_{\text{building blocks}}$$

$$\Big\uparrow \quad \text{four functorial projections [1, Thm 16.2]}$$

$$\underbrace{\chi_{-4}}_{\text{Kronecker character of } \mathbb{Q}(i)}$$

$$\Big\uparrow \quad \text{imaginary-part trace} \ \ \chi_{-4}(n) = \mathrm{Im}(i^n) = \sin(\pi n/2)$$

$$\underbrace{\{-1,\ 0,\ +1\}}_{\text{value set = FTD voxel alphabet (set-level)}}$$

$$\Big\uparrow \quad \text{closure of } \langle i \rangle\text{-orbit under squaring \& modulus}$$

$$\underbrace{i^2 = -1}_{\text{ontological zero-point}}$$

Every arrow above the FTD bridge is established (theorem, observation, or honest conjecture) in [1]. The FTD bridge itself is the physical conjecture that this companion records under `[STRONGLY MOTIVATED CONJECTURE]`. Everything below the bridge is pure mathematics; the only place that requires physical input is the single bit of the exponent-pair selector.

6. Outlook

Two productive lines of work are identified by the descent chain.

Closing the bridge. If a derivation of the exponent pair $(2,3)$ from χ_{-4}-internal data can be made rigorous — the candidate framing in [1, §16.5, Conjecture (Sym^2–Sym^3)] is the most natural starting point — the FTD bridge would become a single-arrow theorem rather than a two-arrow conjecture. This would not prove the empirical match to (α^{-1}, N_c) (that remains an experimental fact about the universe), but it would localise the conjectural content to a single physical assertion: that the universe's electroweak and strong sectors are built from a $\mathbb{Z}[i]$-substrate.

Extending the bridge. The current FTD identifications cover α and N_c. The Standard Model has additional dimensionless constants ($\sin^2\theta_W$, m_μ/m_e, m_τ/m_e, PMNS angles, etc.) which the FTD framework currently parameterises without first-principles derivation. The χ_{-4}-unification suggests that further dimensionless invariants may be discoverable as roots of higher-rank master polynomials living in $\mathrm{Sym}^\bullet(H^1(E_{\mathrm{lemn}}))$, or as Hecke L-values of related CM elliptic curves. This is the natural program for extending the bridge from (α^{-1}, N_c) to the full SM.

A second-tier programme suggested by the parallel equianharmonic structure of [1, §16.4] is the identification of the χ_{-3}-side (equianharmonic, $|\mathrm{Aut}| = 6$) as the structural home of the QCD gauge structure, with χ_{-4} (lemniscatic) as the home of the electroweak structure. The Moore Layer Theorem of FTD [2] produces $U(1) \times SU(2) \times SU(3)$ from a 27-block polyhedral decomposition; whether this can be reformulated as the joint action of χ_{-4} (giving $U(1)$ and part of $SU(2)$) and χ_{-3} (giving $SU(3)$ and the rest of $SU(2)$ via the Hurwitz-quaternion intersection $\mathbb{H} \supset \mathbb{Z}[i], \mathbb{Z}[\rho]$) is an open speculative direction worth investigating but not pursued here.

References

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[5] E. P. Wigner, *The Unreasonable Effectiveness of Mathematics in the Natural Sciences*, Communications on Pure and Applied Mathematics **13** (1960), 1–14.